

The Amplifying Effect of Explainability in AI-assisted Decision-making in Groups

Regina de Brito Duarte
INESC-ID
Lisbon, Portugal
Instituto Superior Técnico
University of Lisbon
Lisbon, Portugal
reginaduarte@tecnico.ulisboa.pt

Joana Campos
INESC-ID
Lisbon, Portugal
Instituto Superior Técnico
University of Lisbon
Lisbon, Portugal
joana.campos@inesc-id.pt

Mónica Costa Abreu
Instituto Superior Técnico
University of Lisbon
Lisbon, Portugal
monica.abreu@tecnico.ulisboa.pt

Ana Paiva*
INESC-ID
Lisbon, Portugal
University of Lisbon
Instituto Superior Técnico
Lisbon, Portugal
ana.paiva@inesc-id.pt

Abstract

In the era of artificial intelligence, AI-assisted decision-making has become a common paradigm. Explainable Artificial Intelligence has been one of the more explored factors in improving transparency of AI tools in AI-assisted decision-making, but sometimes with contradictory results. Furthermore, while individual AI-assisted decision-making has garnered substantial investigation, the domain of group AI-assisted decision-making remains notably underexplored. This research presents the first look at the impact of explainability and team composition on AI-assisted decision-making. With a controlled experiment on mushroom edibility classification, with 89 participants, we show that the impact of XAI is more pronounced in decision-making with groups (2-person) than in individual decision-making. Groups rely less on incorrect AI recommendations when explanations are available, but they rely more on incorrect AI recommendations when explanations are absent, compared to individual decision makers. This phenomenon underscores the amplified effect of explainability in AI-assisted decision-making in group settings.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → *Artificial intelligence*.

Keywords

AI-Assisted Decision-Making, XAI, Groups Decision-Making, Human-AI Interaction

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1 Introduction

The widespread presence of Artificial Intelligence (AI) decision support systems in critical fields such as healthcare [10, 47], finance [40], and justice [18, 20] - where AI assists humans by providing recommendations and predictions for specific decision tasks - has propelled significant research in AI-assisted decision-making [5, 9, 30, 67]. While AI systems have demonstrated remarkable capabilities in enhancing decision accuracy in specific tasks (e.g., recidivism prediction [20] or cancer detection [56]), their role in AI-assisted decision-making extends beyond mere precision. They also significantly shape the dynamics of decision-making, where trust in the AI system is the mediator variable between task complexity and user reliance [3, 4].

The dynamic interplay between trust in AI systems and decisions is critical in effectiveness and adoption of AI technologies, as it influences how users rely on or challenge AI-generated recommendations. Consequently, *reliance* becomes a key metric to characterize human-AI trust levels, as it reflects the extent to which users trust and depend on the AI system [51]. Overtime, it can give us an idea of the trust calibration process in a decision-making task assisted by AI, allowing to assess whether users are *over-relying*¹ or *under-relying*² on the AI system.

This trust calibration process involves decision-makers constructing accurate mental models of the AI system's reasoning mechanisms. When these mental models are poorly formed, users are less likely to use the AI system effectively [3, 4]. To help humans create better mental models, several XAI approaches have

¹Users completely trust of the AI system's decision.

²User are under-utilizing the AI system and/or reveal scepticism towards it.

emerged to improve decision accuracy [28], and calibrate *reliance* on AI recommendations [51, 67]. However, the effectiveness of XAI techniques may vary depending on the specific context (e.g., clinical or industrial) and task structure (e.g., individual or group decision making).

In this paper, we explore the impact of XAI in a *group* task, a largely unexplored area that is crucial for improving our understanding of AI's role in team-based decision-making. Group decision-making is a process utilized in various sectors of our society: in the healthcare field with multidisciplinary teams of experts analyzing a patient case [55] or in corporate organizations during strategic planning decisions made by the board of directors [44, 60]. AI systems are being incorporated into these group decision-making processes [13, 44, 62], but there is still limited understanding about how AI assisted decision and its explainability impact these group decisions.

To address this issue, we conducted a user study (N=89) to investigate the impact of AI explanations on group AI-assisted decision-making. Through a quantitative controlled experiment, we explore patterns of decision accuracy, AI reliance, subjective trust, and understandability in AI-assisted decision-making settings, comparing individuals and groups of two subjects performing a task of mushroom edibility classification. Our findings reveal that the presence of AI explanations has a significantly greater impact on AI-assisted decision-making in groups when we look at AI over-reliance. *Over-reliance* is more evident in the absence of explanations when decisions are made by a 2-person group, as opposed to a single person. Conversely, when explanations are provided, the group of decision-makers exhibits less over-reliance compared to the individual decision-maker. In addition, results show that *under-reliance* remains unaffected by the presence of explanations, suggesting that other approaches are needed to mitigate this effect.

Our work contributes to the field of Human-AI interaction research by offering two major contributions. First, we conduct the first quantitative user study that evaluates the effectiveness of XAI in group AI decision-making, exploring how presenting AI explanations affects decision performance, reliance patterns, and understanding of AI. Second, by applying a more robust study protocol we enable a more effective assessment of both over-reliance and under-reliance and compare their differences in individuals and 2-person groups.

2 Related Work

2.1 The Role of XAI in AI-Assisted Decision-Making

XAI has been a key technique in the Artificial Intelligence and Machine Learning research field. It was primarily developed to address the need to explain black-boxes algorithms [49]. In the Human-AI Interaction community, this technique has been used to improve the collaboration between humans and AI systems. For instance, a study focusing on an AI bird identifier app revealed that users sought explanations to grasp the task, comprehend the AI's workings, make informed decisions about when to trust the system, and provide constructive feedback for app development [28].

Within the context of human AI-assisted decision-making, the impact of explanations on decision performance yields conflicting results. On one hand, well-understood explanations, requiring minimal cognitive load, can mitigate over-reliance on AI systems [63]. Additionally, using example-based explanations has proven effective in reducing over-reliance and improving decision-making by fostering a complementary dynamic between AI and the user [11]. Another approach is to incorporate confidence scores into AI recommendations as explanations, which further contributes to improved trust calibration [67]. On the other hand, some studies suggest that explanations emphasizing feature importance may induce over-reliance [5, 11, 16, 65], whereas other types of explanations may have little or no noticeable effect on decision-making performance and trust in the AI [10, 16, 50]. These results indicate that both the nature of the explanations and the characteristics of the decision-making task can significantly affect the impact of explanations on AI-assisted decision-making. Furthermore, explanations can create a placebo effect, where users develop increased trust in the AI system simply because an explanation is provided, regardless of the explanation's accuracy or the AI system's actual reliability [16, 19].

This diversity of findings motivates the research community to explore alternative designs to explain AI recommendations, with the aim of minimizing the challenges of XAI and amplifying its positive impact on the decision-making process. Approaches such as providing a comprehensive narrative as an explanation for complete understanding [22] or offering insights into the pros and cons of each option instead of explaining the rationale behind an AI prediction [39] represent only two examples of innovative strategies to enhance XAI performance in AI-assisted decision-making. Additionally, another potential improvement lies in the use of cognitive forcing functions, which are design strategies aimed at fostering engagement in the decision-making process [9, 21]. This approach compels decision-makers to actively engage with the explanations, addressing the shortcomings associated with presenting explanations. This design method has proven effective in reducing over-reliance and enhancing overall performance and learning.

2.2 Measuring AI Reliance in AI Systems

To evaluate the influence of AI recommendations on decision-making tasks, researchers have concentrated on three primary metrics: *reliance patterns* related to the AI system - which includes over-reliance and under-reliance -, *trust* in the AI system, and individual *decision confidence*³. This evaluation helps to understand the human-AI interaction in decision-making tasks [51]. Furthermore, reliance patterns as behavioral measures can serve as a proxy to infer trust in the AI system. However, there is no standardized method for measuring reliance on the AI system.

Typically, AI reliance is assessed by identifying cases where the final decision matches the AI recommendation [5, 9, 63]. However, this straightforward approach fails to account for situations where the final decision aligns with the AI recommendation not due to reliance, but because of the individual's confidence in their own judgment. For instance, when a decision-maker has a prior belief that aligns with the AI recommendation, the final decision matches

³The former is a behavioural measure, while the other two are self-reported.

the AI recommendation, but the main driver was the user's prior belief.

According to the definition proposed by Schemmer et al.[51], AI reliance happens only when a decision-maker holds an initial belief different from the AI's recommendation and chooses to adopt the AI's advice over their own belief. This definition takes into account the human decision process that occurs before viewing the AI's suggestion. With this framework, over-reliance and under-reliance can be more accurately described and measured. Over-reliance occurs when the decision-maker relies on the AI system despite it providing an incorrect or misleading recommendation. It often arises in contexts that involve lay users or when tasks are excessively complex and require substantial cognitive effort [9, 58, 63]. Conversely, under-reliance tends to occur when users are skeptical of the AI's capabilities or responses and thus the decision-maker does not rely on the AI system even though it offers a correct recommendation. Both extremes of reliance patterns are detrimental, significantly influencing the dynamic between the human and the AI assistant, and are closely interrelated with decision performance [53]: over-reliance leads to errors by accepting incorrect AI outputs, while under-reliance arises from rejecting helpful AI input, both ultimately reducing performance quality.

2.3 Decision-Making in Groups

Group decision-making has been the subject of extensive research, with a focus on factors such as group composition, communication patterns, and the nature of the decision task [27, 31, 57, 66]. Collaborative efforts in group decision-making have been shown to mitigate individual biases, enhance outcomes, and often result in superior performance compared to individual endeavors [6]. Although groups offer several benefits, they are not without challenges. Phenomena such as groupthink [24], where individuals in the group align with the majority opinion at the expense of critical analysis, and internal polarization [42, 52], where group opinions and beliefs become more extreme and polarized compared to those held by individual members, exemplify these difficulties and demonstrate the nuanced nature of group decision-making.

Research on whether groups surpass individuals in decision-making yields mixed results. Some studies emphasize significant advantages of groups [37], while others highlight a process loss phenomenon [27]. The comparison between individual and group decision-making is nuanced by factors such as task complexity [1], performance disparities among individuals in the group [2], and the alignment of individuals' subjective confidence in their accuracy [29]. However, social influence is a common factor in most group decision-making settings that can undermine its performance [34]. In fact, the presence of social influences introduces more noise in group decision-making compared to a simple statistical estimate of individual independent decisions [26].

While considerable attention has been devoted to understanding the dynamics of group decision-making, the exploration of group AI-assisted decision-making is a more recent development. This evolution aligns with significant advancements in AI decision-making and extensive investigations into individual AI decision-making. Munyaka et al. [41] have studied the differences between human-human-AI teams and human-AI-AI teams, considering different

decision-making styles. Their research shows that perceived team efficacy and performance are higher in human-AI-AI teams when the AI partner employs an autocratic decision-making style compared to human-human-AI teams with an autocratic human partner. Additionally, an exploratory study examined interactions between two humans and one AI agent in an essay-ranking task [68]. The findings suggest that humans were not always favorable toward sharing equal voting rights with the AI agent, and that the AI agent, although competent in its recommendations, exhibited rigidity in its interactions. To our knowledge, only one study has directly compared individual to group decision-making within an AI-assisted context. Chiang et al. conducted a controlled user experiment on an online, AI-assisted recidivism risk assessment task. The researchers manipulated the number of participants assigned to complete the task, with participants working either individually or in groups of three. In the group condition, participants could communicate in a chat room to reach a consensus. Preliminary findings from this study indicate that decision accuracy remains unaffected in group settings compared to the individual setting, yet there is an observed increase in reliance on AI recommendations regardless of AI correctness [12]. This study highlights the potential differences in outcomes within AI-assisted decision-making when decisions are made by groups rather than individuals. However, further exploration is needed to examine the role of AI explainability as a potential influencing factor in the decision-making process. Additionally, incorporating users' pre-existing beliefs into the measurement of reliance could provide deeper insights, as discussed in Sec. 2.2.

In our research, we delve deeper into the effects of AI assistants on group decision-making by examining the influence of presenting AI explanations on decision performance, AI reliance patterns and AI understandability. This study allows us not only to investigate underexplored effects of AI-assisted decision-making in groups such as the reliance patterns (over-reliance and under-reliance) and the decision performance, but also to contribute to the research on the effects of XAI in AI-assisted decision-making.

3 User Study

To explore how AI contributes to team-based decision-making and how AI explanations affect groups' decisions versus individuals' decisions, we focused on two research questions:

- [RQ1] In AI-assisted decision-making scenarios, what impact do AI explanations (XAI) have on the **performance** of decision-making by two-person teams?
- [RQ2] How **reliance** on AI differs between decisions made by individuals alone versus decisions made by two people working together as a pair?

To address our research questions, we carried out an experimental user study. The experiment involved a task of mushroom edibility classification and manipulated two variables: the inclusion of explanations during decision-making (**XAI**) and the number of human decision-makers (**Team Composition**). The research employed a 2x2 between-subjects design, with each participant experiencing a unique condition from the XAI and Team Composition factors.

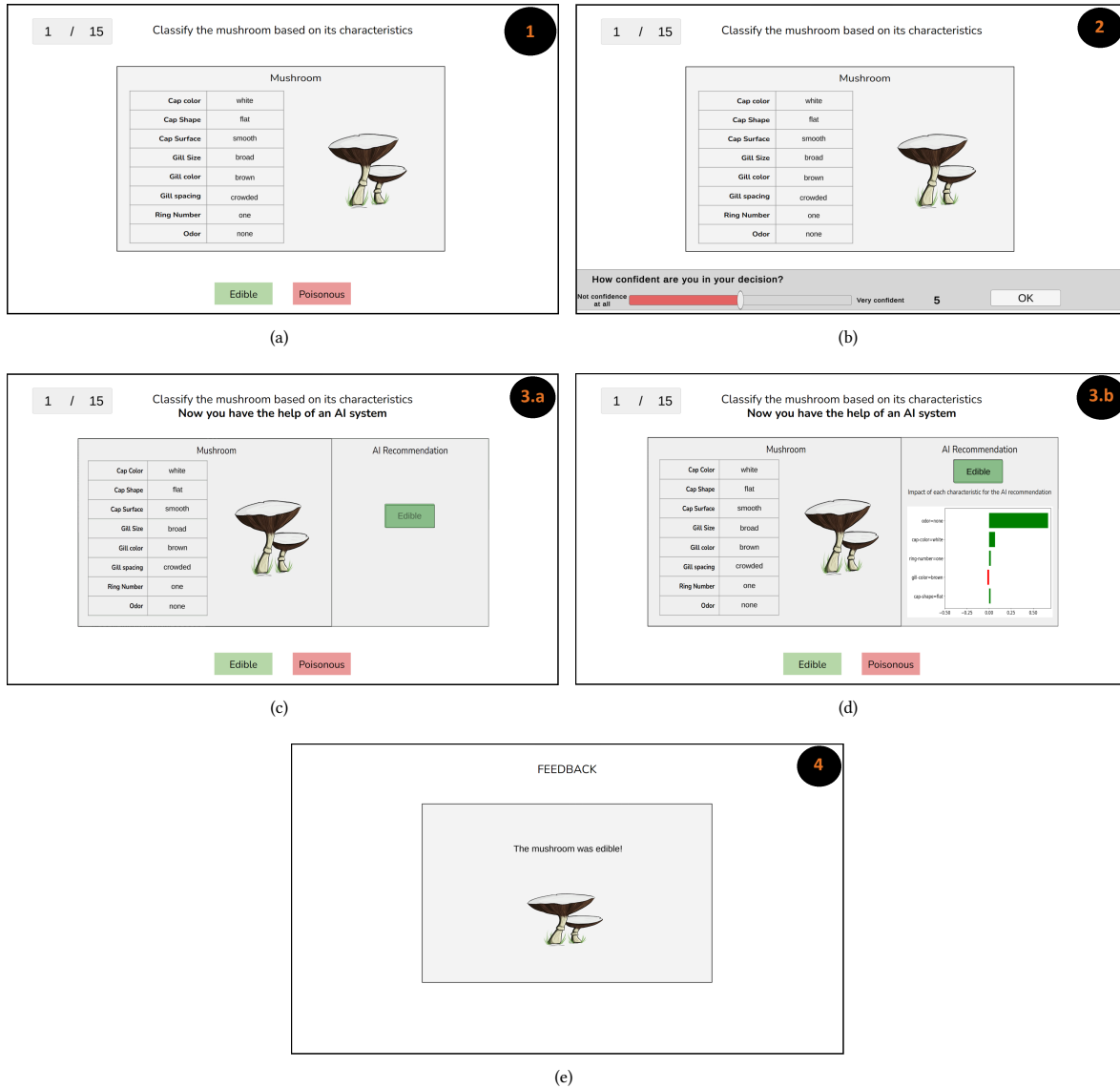


Figure 1: The mushroom edibility task. Subjects start the task by determining the edibility of a mushroom presented to them (whether it's edible or poisonous). They can see their eight characteristics and an visual representation (a). Subsequently, a pop-up question appears in the screen asking about their confidence in the decision they made (b). Subjects, then, make a second decision about the same mushroom but now with the help of an AI with its recommendation (c) or with a recommendation and its explanation (d). Finally, subjects receive feedback on the real edibility of the mushroom (e).

3.1 Task Description

The task used in our study consisted in evaluating the edibility of a given mushroom (either edible or poisonous) based on a visual representation of the mushroom and their eight characteristics: *Odor*, *Cap Color*, *Cap Shape*, *Cap Surface*, *Gill Color*, *Gill Size*, *Gill Spacing*, and *Ring Number* (see Fig. 1(a)). These characteristics were categorical, with each having between 2 possible options (e.g., gill size) and 12 possible options (e.g., gill color). The mushroom edibility assessment was done in four steps.

In the first step (Fig. 1(a)), participants assess the mushroom as either edible or poisonous based solely on its visual representation and a description of its eight characteristics. Following that, participants are asked to rate their confidence in their decision (Figure 1(b)). In the third step, they reevaluate the same mushroom, this time with additional information: an AI recommendation on the mushroom's edibility without (Figure 1(c)) and with explanations (Figure 1(d)), depending on assigned the condition. Ultimately, participants received feedback regarding the actual edibility of

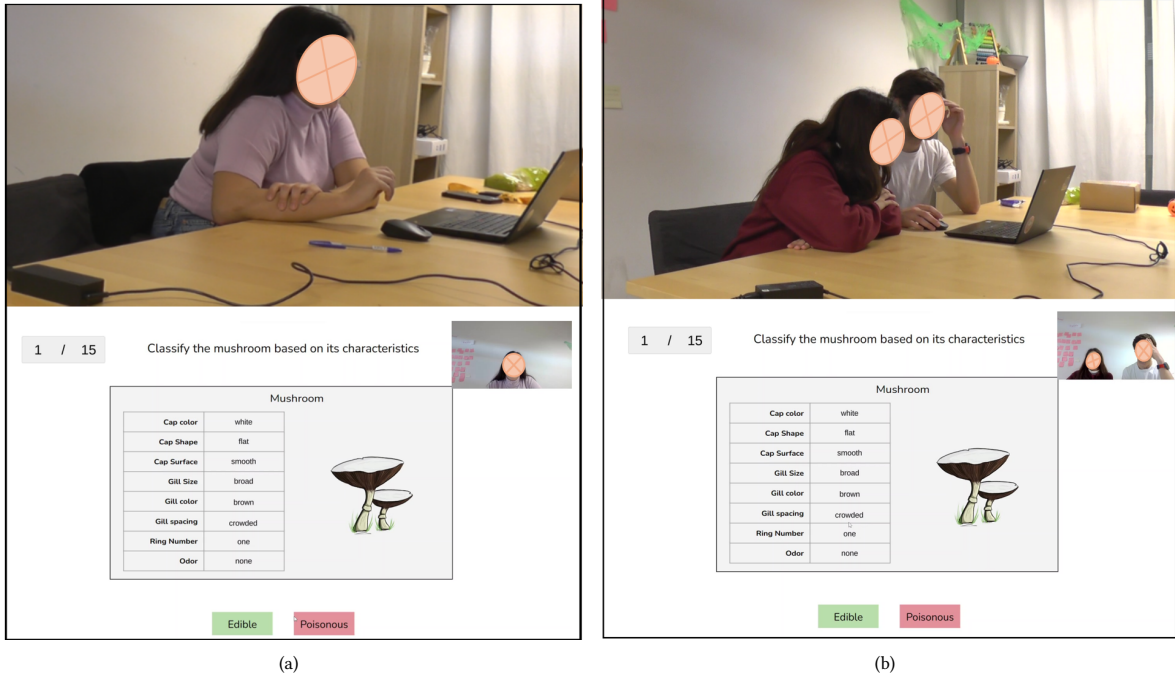


Figure 2: Participants executing the required task: a) A participant in the *Individual Condition* performing the task. b) A group of two subjects in the *Group Condition* collaboratively completing the task. The lower images depict the participants' view of the screen and what the computer's webcam recorded, which participants did not see during the task.

the mushroom (Figure 1(e)). The AI algorithm used to generate recommendations is a decision tree based on a tabular dataset of mushroom characteristics [46] and an accuracy on the test set of 96%.

This task required participants to make two consecutive decisions about the mushroom's edibility: first, without any AI assistance, and then with the aid of an AI assistant. This experimental design decision of requiring participants to make two sequential judgments, first without and then with the AI recommendation, is critical to measure AI reliance, as defined in [51]. Further details on this definition can be found in Sec. 3.6.

The reason for selecting this task is based on two primary considerations. First, it incorporates a psychological element of risk, though not real, as participants might mistakenly classify a poisonous mushroom as edible. This perceived risk is crucial for assessing trust and reliance on the AI. Without such an element, evaluating AI trust and reliance could be less effective [23, 38]. Second, this task is universally understandable, even for individuals unfamiliar with mushroom edibility, as it aligns with basic common sense or misconceptions about mushrooms. Hence, this task accommodates different levels of expertise among participants. Furthermore, mushroom picking can be a group activity, making it a good task for group-based decision-making and is often used for research purposes in AI-assisted decision-making scenarios [16, 32, 33].

3.2 Experimental Conditions

We manipulated two factors to understand the effect of explanations on group-AI assisted decision-making: the number of human subjects performing the task, denoted as **Team Composition**, and the presence of an explanation for the AI prediction, denoted as **XAI**.

Team Composition. We varied the number of human decision-makers involved in the task, creating two distinct conditions. In the *Individual Condition*, the mushroom edibility assessment task was performed by a single participant, who evaluated mushrooms, considered AI predictions, and made decisions independently. In the *Group Condition*, the task was conducted by two subjects working collaboratively. Subjects were encouraged to engage in discussions to reach a consensus before making their joint decision.

Presence of Explanations (XAI). Our manipulation involved presenting an explanation of the AI recommendation. In the baseline condition, *Without XAI Condition*, no explanations were provided. Subjects had access solely to mushroom characteristics and AI predictions (see Figure 1(c)). In the *With XAI* condition, subjects could also view an explanation of the AI recommendation (see Figure 1(d)). This explanation showed the importance of each mushroom characteristic for the decision and was based on the *LIME* algorithm [16, 48]. The choice of the type of explanations was based on accessibility to lay users. People without expertise in mycology are unfamiliar with mushroom species, the guidelines for their edibility, and lack an understanding of mushrooms themselves.

Therefore, the provided explanations must be straightforward and not dependent on any prior mushroom knowledge. In this context, feature importance explanations are more appropriate than counterfactual or example-based explanations.

3.3 Experimental Setting

The entire experience was conducted in person to facilitate interaction and discussions among subjects in the *Group Condition*. The studies were carried out in a dedicated room at the University. At arrival participants were directed to a laptop computer where the mushroom classification task was running. The task was developed using Unity and was running locally.

Two cameras were used to record the participants. The first was the laptop's built-in webcam, and the second was positioned diagonally to capture their movements. Figure 2 displays the setup from both the webcam and the secondary camera viewpoints. For the *Group Condition*, participants volunteered in groups of two. Therefore, they knew each other and had similar backgrounds and demographics. The study protocol was approved by the University's Ethics Committee.

3.4 Experimental Procedure

Each session started with a short introduction to the experiment. Participants were informed about their rights and signed a consent form for behavioural and video data collection. Then a concise overview of the task was presented to the participants.

The participants initiated the experiment by familiarizing themselves with mushroom characteristics and undergoing a task tutorial consisting of two rounds. Upon completing the tutorial, they proceeded to the primary task. The primary task involved conducting fifteen rounds of the mushroom edibility classification task. Each round consisted in the mushroom edibility task described in Sec. 3.1. The order of appearance for the 15 mushrooms was the same for all participants, with the AI making correct predictions in twelve out of the fifteen mushrooms. The three incorrect AI predictions occurred in the 5th, 10th, and 13th positions in the sequence. The decision to allow 3 AI misclassifications (corresponding to 80% perceived accuracy) aimed to balance the need for measurable errors to assess over-reliance and to maintain a realistic, user-supportive AI algorithm. In the *Group Condition*, subjects were encouraged to engage in collaborative discussions and reach a consensus in each round, before making a decision.

At the conclusion of the task, participants were asked to complete a questionnaire. This questionnaire, administered individually including those in the *Group Condition*, included demographic inquiries and solicited feedback on the experiment for the subjective measures described in Sec. 3.6. Each participant received a cinema ticket voucher in the value of €6.5 for their participation.

3.5 Participants

A total of 61 sessions were conducted with 92 participants recruited on-site at the University. However, only 59 of these sessions were deemed valid following data quality checks⁴, which accounts for 89 participants. Among the participants, 44 identified as female, and 45 identified as male. The predominant age range was between 18 and

25, with only 13 participants falling into an older age bracket. The number of experiments conducted per condition is outlined in Table 1. From a decision-wise perspective, a total of 885 decisions were made (59 sessions multiplied by 15 rounds). The flow of decisions and their frequencies is illustrated in Figure 3.

3.6 Measures

The objective of this user experiment is to investigate the impact of explainability in AI-assisted group decision-making, specifically examining its influence on decision performance and AI reliance. The study places a primary emphasis on measures related to AI reliance, trust in AI, and decision performance, assessed through both behavioral and subjective dimensions.

3.6.1 Behavior Measures. When assessing behavioral measures, we focus on subjects' actions and behaviors. We analyze these behaviors from a subject-wise perspective: each metric is examined based on a composite of all decisions made by an individual/group. The following measures are defined for this perspective:

Decision Accuracy. Refers to the accuracy of each decision. Subject-wise, is the rate of correct decisions made by each subject (or group). The accuracy of both the initial human decision and the final decision with AI assistance was measured.

Decision Confidence. Subject's confidence in each initial decision was quantified on a scale from 0 (not confident at all) to 10 (very confident). Subject-wise, the metric measured the average confidence for each subject.

Decision Time. The time spent on each decision, both the initial human decision and the final one, was recorded. In the subject-wise perspective, each subject was associated with two variables: the mean time spent on the initial human decision and the mean time spent on the final decision.

Reliance Measures. To gain a nuanced understanding of reliance patterns, we assessed *AI reliance*, but also *AI under-reliance*, and *AI over-reliance* according to [51]. Therefore, we characterize the reliance metrics as follows:

- **AI Reliance Rate:** AI reliance refers to relying on the AI recommendation when it is against one's judgment. Hence, a final decision is considered reliant if it matches the AI prediction, but differs from the initial human decision. The AI reliance rate is defined as the ratio of reliance cases to all cases within the reliance domain - all the cases that could be considered reliance. Figure 3 has a diagram that describes the reliance domains.
- **AI Over-reliance Rate:** AI over-reliance refers to erroneously relying on the AI recommendation when it is against one's judgment. Hence, a final decision is considered over-reliant if it matches the AI prediction; differs from the initial human decision and the AI prediction is wrong. The AI over-reliance rate is defined as the ratio of all over-reliance cases to all cases within the over-reliance domain,

⁴Two sessions were excluded due to incomplete data records.

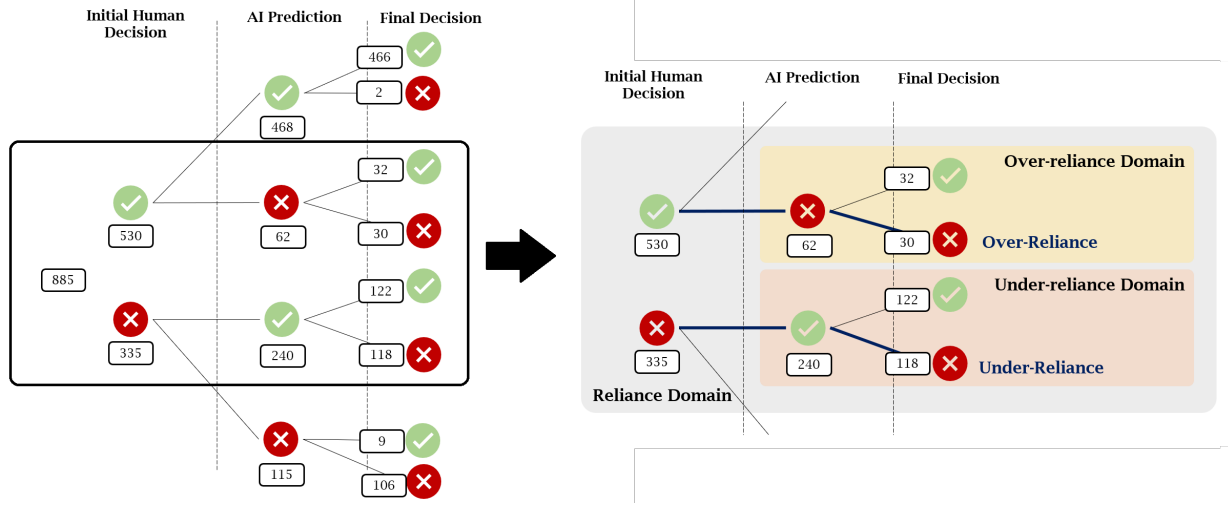


Figure 3: The correctness flow diagram illustrates various combinations of Initial Human Decision, AI Prediction, and Final Decision. On the left, the distribution of decisions in each combination is presented. On the right, zoom in on the cases within the reliance domain. A decision falls into the reliance domain only if the initial human decision differs from the AI prediction; otherwise, it is indistinguishable whether the final decision was made relying on the AI or if it was self-reliance. From the cases that are in the reliance domain, only the ones in which the final decision is equal to the AI prediction are considered reliance. Concerning over-reliance, a decision can be categorized as such only if it falls within the reliance domain and the AI prediction is incorrect; otherwise, the final decision cannot erroneously be reliant on the AI. The under-reliance domain follows a similar pattern, but when the IA has a correct prediction. The reliance domains are important to limit the scope of possible cases of reliance, over-reliance and under-reliance.

where the initial human decision differs from the AI prediction, and the AI prediction is wrong (see Figure 3 for more detail). Note that AI over-reliance is a subpart of AI reliance: over-reliance regards the cases where one's relies on the AI recommendation erroneously, while AI reliance refers to cases where one's relies on IA recommendation regardless of AI correctness.

- **AI Under-reliance Rate:** AI under-reliance refers to erroneously self-relying on one's own judgment against a correct AI recommendation. Hence, a final decision is considered under-reliant if it differs from the AI prediction; matches the initial human decision and the AI prediction is correct. The AI under-reliance rate is defined as the ratio of all under-reliance cases to all cases within the under-reliance domain, where the initial human decision differs from the AI prediction, and the AI prediction is correct (see Figure 3 for more detail).

3.6.2 Subjective Measures. In the end of the task, we asked all subjects to fill out a questionnaire individually regarding the task. It serves to measure their subjective trust and understandability about the AI. We also added two open-ended questions to the questionnaire, inquiring about the reasoning behind their decisions and the manner in which the AI recommendations were incorporated into their choices.

Subjective Trust. We assessed subjective trust in the AI using the Multi-Dimensional Measure of Trust [36]. This measure captures indirect subjective trust by evaluating participants'

perceptions of eight traits associated with the AI system: Reliable, Predictable, Consistent, Skilled, Capable, Competent, Precise, and Transparent. Participants rated their responses on a scale ranging from 0 (not at all) to 7 (very), and then the scores were averaged.

Subjective Understandability we assessed the system's understandability using three items ("I understood how the AI system works"; "The information presented on the screen to decide the edibleness had sufficient details" and "From the information presented on the screen, I understood how the system made its decisions"). These items were also rated on a scale ranging from 0 to 7 and averaged.

4 Results

The results were analyzed using ANOVA statistics and all the interaction effects between team composition and XAI and the main effects are reported. Simple effects are reported whenever necessary. Descriptive statistics are reported in the reduced format ($Mean \pm sd$).

4.1 Decision-Making Performance

4.1.1 Decision Accuracy. The interaction between Team Composition and XAI showed no statistically significant effect on initial decision accuracy ($F(1, 55) = 0.075, p = 0.785$). Similarly, the Team Composition factor alone did not significantly impact initial decision accuracy ($F(1, 55) = 0.710, p = 0.403$). In contrast, the XAI

Conditions	Individually	In Group	Total
Without XAI	15	14 (28 participants)	29 (43 participants)
With XAI	14	16 (32 participants)	30 (46 participants)
Total	29	30 (60 participants)	59 (89 participants)

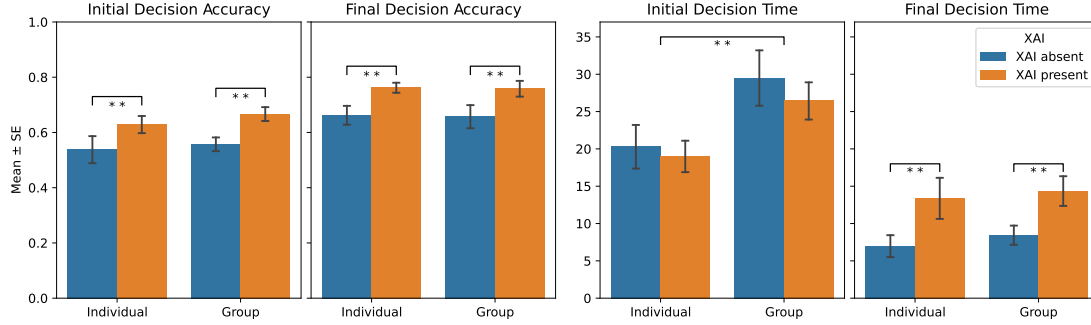
Table 1: Number of experiments (and participants) per condition.

Figure 4: Decision accuracy and average decision time for the initial and final decisions considering XAI and Team Composition factors. The two plots on the left-hand side display initial and final decision accuracy rates, and on the right-hand side two plots of the time spent in both the initial and final decisions. The error bars represent the standard error. $*p < 0.05$, $* * p < 0.01$, $* * * p < 0.001$

factor had a statistically significant effect on initial decision accuracy ($F(1, 55) = 8.268, p = 0.005$). Participants provided with explanations achieved a initial decision accuracy rate of $0.65(\pm 0.11)$, whereas those who received only AI recommendations initially had a decision accuracy of $0.55(\pm 0.15)$. Refer to figure 4 for a visual representation.

Regarding the final decision accuracy, there was no statistically significant interaction effect between Team Composition and XAI ($F(1, 55) = 0.001, p = 0.981$). The final decision accuracy rate for the participants in the Group condition and participants in the Individual Condition was $0.711(\pm 0.143)$ and $0.710(\pm 0.116)$, respectively. The ANOVA F-test result suggests that this difference is not statistically significant ($F(1, 55) = 0.019, p = 0.892$). However, similarly to the initial decision accuracy rate, the final decision accuracy of participants who had access to explanations was higher than that of participants without explanations (0.76 ± 0.094 and 0.66 ± 0.142 , respectively). This difference is statistically significant ($F(1, 55) = 9.98, p = 0.003$).

By comparing the accuracy of both the initial and final decisions, we observe that the final decision, which includes AI recommendations, yields a higher accuracy (0.71 ± 0.13) compared to the initial decision's accuracy (0.60 ± 0.14). This difference is statistically significant ($F(1, 55) = 41.2, p < 0.001$).

4.1.2 Decision Confidence and Decision Time. There was no statistically significant interaction effect between Team Composition and XAI on the mean decision confidence for each subject ($F(1, 55) = 0.231, p = 0.632$). The average confidence after making the initial decision was very similar for participants with explanations (6.02 ± 1.76) and those without explanations (6.11 ± 1.46) - There was no statistically significant effect of the XAI factor

($F(1, 55) = 0.079, p = 0.78$). Concerning the impact of team composition, subjects in the group condition exhibited higher confidence (6.37 ± 1.21) than subjects in the individual condition (5.75 ± 1.90), although this difference was not statistically significant ($F(1, 55) = 2.2, p = 0.14$).

Regarding the mean time spent making decisions, the only significant factor that influenced the time spent on the initial decision was the Team Composition. As anticipated, subjects in the *Group condition* invested more time in making the initial decision (27.86 seconds ± 11.83) compared to subjects in the *Individual condition* (19.66 ± 9.67) ($F(1, 55) = 8.502, p = 0.005$). However, when examining the time spent on the final decision, team composition did not show a statistically significant effect ($F(1, 55) = 0.393, p = 0.533$), while the XAI factor did. Subjects with access to explanations of the AI system took more time to make the decision (13.90 ± 8.96) compared to those who only had the AI recommendation (7.68 ± 5.24). This difference was statistically significant ($F(1, 55) = 9.97, p = 0.003$). Figure 4 shows the initial and final average decision time for each factor.

4.2 AI Reliance

Table 2 presents the ANOVA results for each reliance measure.

4.2.1 Reliance. When assessing the overall level of reliance, no discernible impact of XAI or team composition factors was observed – neither factor had a statistically significant effect on reliance rates. Nevertheless, an interaction effect was identified. As depicted in Figure 5, participants in the Group Condition who had access to explanations exhibited lower reliance rates (0.33 ± 0.25) on the system compared to those without access to explanations (0.59 ± 0.28). This disparity although noticeable is not statistically significant

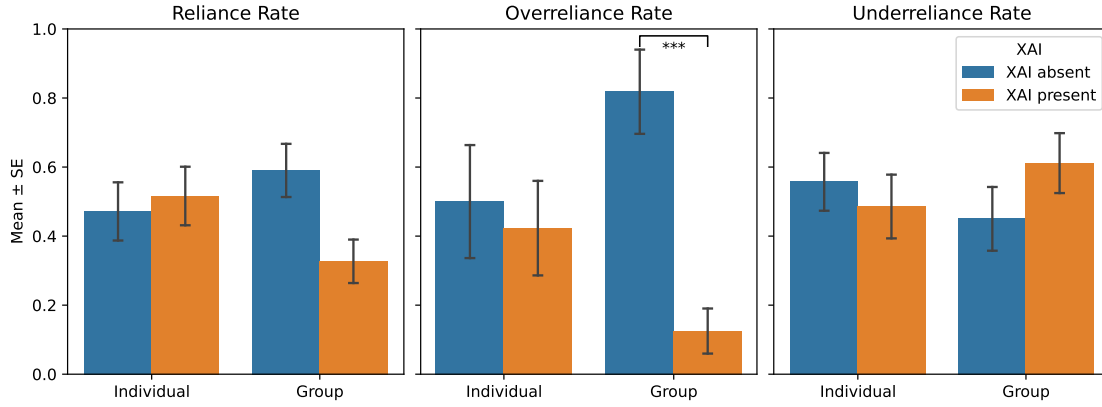


Figure 5: AI Reliance rates considering XAI and Team Composition factors. In the left is a graphical visualization of the reliance rates, in the middle a visualization of the over-reliance rates, and in the right an illustration of the under-reliance rates. The error bars represent the standard error.

Reliance Measure	Factor	$F(1, 55)$	p-value	η^2
Reliance	XAI	2.00	0.16	0.035
	Team Composition	0.208	0.65	0.004
	XAI * Team Composition	3.97	0.05	0.067
Over reliance	XAI	9.523	0.004**	0.192
	Team Composition	0.006	0.94	0.00
	XAI * Team Composition	6.098	0.018*	0.132
Under reliance	XAI	0.256	0.615	0.005
	Team Composition	0.011	0.917	0.00
	XAI * Team Composition	1.720	0.195	0.03

Table 2: ANOVA main effects for the three reliance measures: AI reliance, AI Over-reliance and AI Under-reliance. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

when correcting the p-value for multiple tests using the bonferroni method ⁵.

4.2.2 Over Reliance. There was an interaction effect between Team Composition and XAI in the over reliance rates. When explanations are present, participants that do the task in group have lower rates of over reliance (0.125 ± 0.23) compared to participants that do the task individually (0.42 ± 0.49). Moreover, when the decision is made in group, the over reliance rate when explanations are present is lower (0.125 ± 0.23) compared to when the explanations are absent (0.82 ± 0.40). Figure 5 illustrate this interaction. Looking at the other factors, we found no significantly effect on team composition, but we encountered a main effect in XAI. Analysing the simple effect of the XAI factor, we found that the XAI factor is only statistically significant in the group condition ($p < 0.001$) ⁶.

4.2.3 Under Reliance. The under-reliance rates for each factor are illustrated in figure 5. The ANOVA test indicated that the observed differences were not statistically significant (refer to Table 2). Nor

⁵Bonferroni correction applied to the ANOVA F-tests of the three reliance metrics, where the new p-value threshold is 0.016.

⁶Despite the violation of the equal variance assumption for the over reliance measure, the Kruskal-Wallis non-parametric test yields comparable results ($p < 0.001$).

the XAI nor the team composition had a statistically significant effect on under-reliance.

4.2.4 AI Reliance in groups. To investigate the relationship between time to reach consensus and reliance outcomes in groups, we conducted a correlation analysis, using the Pearson correlation coefficient, between the mean time spent in the final decision and reliance rate of each reliance measure. The reliance rate was negatively correlated with the time spent in the final decision, but it was not statistically significant, $r(30) = -0.29, p = 0.12$. In contrast, a significant negative correlation was observed between the over-reliance rate and the time spent in the final decision, $r(30) = -0.65, p < 0.001$. This indicates that higher over-reliance rates were associated with shorter average final decision times. Finally, there was no significant correlation between the under-reliance rate and the time spent in the final decision, $r(30) = 0.06, p = 0.75$.

4.3 Subjective Measures

Participants in the *Group Condition* reported similar levels of subjective trust in the AI (5.36 ± 0.74) as participants in the *Individual Condition* (5.40 ± 0.98). However, participants who had access to the explanation reported significantly higher trust (5.64 ± 0.66) than

those who only saw the AI recommendation (5.09 ± 0.88) ($F(1, 88) = 12.711, p < 0.001$). There was no statistically significant effect of the interaction between the two factors ($F(1, 88) = 1.16, p = 0.285$).

Regarding subjective understandability, a similar pattern is observed. The Team Composition factor does not statistically significantly affect the subjective understanding of AI (4.32 ± 1.77) for Individual Condition and (4.51 ± 1.39) for Group Condition. Only the presence of an explanation is statistically significant. Participants in the With XAI condition reported higher values of understandability (5.07 ± 1.29) compared to participants in the Without XAI condition (3.82 ± 1.50); ($F(1, 88) = 14.906, p < 0.001$). The interaction effect was also not statistically significant ($F(1, 88) = 0.055, p = 0.815$).

4.4 Exploratory Qualitative Analysis

To gain deeper insights into the differences between conditions and the effects of XAI and team composition on AI-human interactions, we analyzed participants' responses to open-ended questions (see Sec. 3.6.2) that explored their reasoning processes and use of AI. We identified themes that participants frequently mentioned, which offer insights into the quantitative results.

4.4.1 Use AI to learn the task. Several participants reported using the AI suggestions to identify patterns and understand how to classify mushrooms as edible or poisonous. Those with access to explanations noted that they relied on the feature importance information to determine the most critical features, using this knowledge in subsequent rounds. For instance, P.TX8⁷ remarked, "We learned how the AI classified them and did the same [...] It taught us that color and odor have more weight in the decision." Similarly, participant P.TX11 stated, "Learned what the AI used as parameters, a lot." Participants appeared to use the information provided in the explanations to enhance their mental model of the task and make more informed decisions in subsequent rounds. This strategy was more commonly reported in groups than in individuals. Notably, 50% of participants in pairs with access to explanations reported using the AI to learn the task, compared to only 27% of individual participants under the same conditions. This might explain why groups rely less on incorrect AI recommendations when explanations are available.

In contrast, participants without access to explanations also attempted to identify patterns in mushroom characteristics but did not use the AI recommendations for this purpose. For example, a participant in session P.T9 explained, "Basically, we based ourselves on previous results, compared different mushrooms with their respective characteristics, and tried to look for a pattern." This process was more challenging without explanations, as information was not distilled and participants had to rely on their own memory of visual characteristics and system's feedback to identify patterns.

4.4.2 Use AI to confirm the initial decision. The mushroom edibility task involves two choices: an initial decision without an AI recommendation and a final decision with access to AI recommendation, along with an explanation in the With XAI scenario. Many participants indicated that they used AI recommendations to validate their

first belief. A participant in session P.T4 mentioned, "And then, usually still followed our instinct, not focusing heavily on AI, essentially confirming what we thought [...]", while P.T8 noted, "The AI served more as reinforcement for the decision[...]" and P.X9 stated, "If we had the same opinion, then I would just stick with my original choice." This pattern was observed across all conditions and underscores the confirmation bias inherent in AI-assisted decision-making. This bias suggests that when users perceive AI recommendations that align with their pre-existing beliefs, they are more inclined to accept them without thorough deliberation [25, 59, 64].

Participants reported increased caution and deliberation only when the initial decision clashed with the AI's recommendation, indicating a discrepancy. For instance, a participant from P.X1 commented "When the prediction aligns with my prediction, I agreed with AI. When they did not align with my predictions I decided based on what I thought but with the information (graphic) that the AI gave." Similarly, a participant from session P.TX2 remarked "If the majority of answers tend towards one answer, I will rethink the decision if it is contrary to my answer." This approach of reconsidering their decision when the AI's recommendation was misaligned was more frequently observed among participants in the With XAI condition.

4.4.3 Use AI to guide their final decision. Many participants reported relying on AI suggestions, often favoring the AI's recommendations over their own judgment due to a lack of confidence in their own knowledge. As a result, the AI's recommendation frequently determined the final decisions in each round. For example, participant P.10 shared, "[...] As I doubt myself a lot, I almost always followed what the AI system told me." Similarly, participant P.X14 stated, "I followed the AI recommendations all the time except once, even when it differed from my initial decision, as I assumed it must have a higher level of knowledge about mushroom edibility." This reliance was especially noticeable when participants were uncertain about their initial decisions. For instance, participant P.3 remarked, "When I wasn't so sure about my decision, I trusted the AI more than my intuition because it is better at noticing subtle patterns and was trained for this."

While this pattern was observed across all conditions, it was less common among participants in pairs with access to explanations. Only two participants in this condition reported relying primarily on the AI for their decisions, compared to eight groups in the condition without explanations who reported the AI as their main guide. In the individual condition, the number of participants that reported using the AI recommendations as the main driver for the decision was similar in both with and without explanations settings, with four participants in each reporting the AI as the primary influence. The tendency to use AI as the primary influence to the final decision was most pronounced in group settings without explanations. This reported behavior provides an explanation for the increase in over-reliance observed in groups without explanations.

4.4.4 Calibrate trust in AI through rounds. Participants reported changes in their confidence levels toward the AI system throughout the experiment. Initially, some fully trusted the AI, but their confidence decreased after encountering incorrect suggestions. For instance, participant P.T5 remarked, "In the beginning, I was trusting 100%, but then I trusted 50%" upon noticing errors. Similarly, participant P.T7 stated, "[...] However, when it failed for the first time,

⁷Participant identifiers are in the format P.TXN, where T indicates the Group condition, X indicates the With XAI condition, and N is the participant's unique number for each conditions combination

we started to ignore it when there was a very big similarity between mushrooms.". Conversely, other participants found their trust in the AI increased as it consistently provided correct recommendations. Participant P.T2 observed, *"The more cases the AI was right, and I was wrong, the more the AI skewed my decision to one side or the other."* Likewise, participant P.T13 shared, *"After I saw that I could trust the AI (after 5 decisions), I decided to follow it."* These statements suggest that participants calibrated their trust dynamically based on the AI's performance during the task. Notably, this trust calibration was most frequently reported by participants doing the task in group without access to explanations.

5 Discussion

This study explores the impact of providing explanations of AI recommendations in an AI-assisted decision-making scenario involving varying team compositions: individuals making decisions with AI assistance and groups of two subjects collaboratively making decisions with the help of an AI assistant. The results indicate a positive influence of explanations on subjective trust and understandability, as well as a reduction in over-reliance on the AI system. Decision makers express higher trust in the AI system when explanations are present and exhibit less over-reliance in such scenarios. However, there is no discernible effect of explanations on under-reliance rates; subjects continue to demonstrate reluctance to rely on the system even in the presence of explanations. Concerning the Team Composition factor, we did not find evidences that decision-making performance or AI trust differs between groups, **but the effect of XAI is amplified in group decision-making**. In instances where the decision task is undertaken by a pair of individuals, over-reliance decreases when explanations are present, and conversely, it increases significantly when only the AI recommendation is provided (see figure 5). The rest of this section provides an in-depth discussion of these findings in the context of our two research questions ([RQ1,RQ2]), the current state of the art in the field, and possible impacts for the research community. In the end, the limitations of this study are given alongside future research directions.

5.1 Decision Accuracy, Reliance and Trust

Decision accuracy was unaffected by team composition; individuals demonstrated the same level of decision performance as groups in the mushroom edibility assessment task [RQ1]. These findings are expected as they align with the initial study that delved into AI-assisted decision-making in groups, revealing equivalent decision accuracy between group-AI and individual decision-makers [12].

Subjective trust and the subjective understandability of the AI produced similar outcomes in our study, with group decision-making having no significant impact on these subjective perceptions. These metrics exhibited changes only when explanations were provided: the presence of explanations led to increased subjective trust and understandability, as well as more accurate decision-making [RQ1].

It is interesting to note that these observed patterns in decision accuracy and subjective trust do not directly align with the AI reliance metric but do correlate with each other. This implies that the trust reported by participants in the questionnaire does not directly correspond to their reliance behavior while using the AI

system. This discrepancy between reported measures and behavioral measures is a phenomenon commonly observed in studies where reported metrics do not align with behavioral metrics [11, 16]. In this context, reported trust and understandability appear to be influenced solely by the presence of explanations, mirroring the impact on decision accuracy. One potential explanation for these results is that reported metrics are associated with participants' decision performance. Participants received feedback after each task, and in the presence of explanations, they comprehended the information, constructing a more accurate mental model of the AI. Consequently, they make more accurate decisions. This process likely contributes to a heightened perception of the system's understandability and an increased level of reported trust. However, this realization seems to occur only in the aftermath of the experiment, when participants can reflect on the results and form a mental representation of the AI system's quality.

We cannot determine whether it is the presence of explanations itself or the higher final decision accuracy when explanations were provided that impacts subjective perceptions. Future research should explore these questions in greater detail. Nevertheless, our findings strongly indicate that reliance behavior does not align with reported trust, at least within the context of one-shot studies. It remains an open question whether this relationship could change in a scenario where subjects use the system over an extended period of time.

5.2 Explanations in Group decision-making

Explanations play a pivotal role in the realm of AI-assisted decision-making and are widely recognized as a crucial factor in enhancing transparency and understandability, thereby influencing the calibration of trust and reliance in AI systems [28, 48]. Our study's findings strongly support the argument that explanations can positively impact human-AI performance by enhancing decision accuracy, aligning with previous research [11]. Furthermore, our results provide evidence that explainability can effectively mitigate over-reliance. This aligns with the insights presented by Vasconcelos et al. [63], who argues that explanations can reduce over-reliance, particularly when they are presented in an easily understandable manner. In our study, the majority of participants lacked expertise in the task of assessing mushroom edibility. However, the explanations provided not only enhanced their understanding of the task but were also presented in a manner that was easy to comprehend and follow. The qualitative exploratory analysis (in Sec. 4.4) supports this argument. Participants in the *With XAI condition* reported that the AI played a crucial role in helping them to understand the key factors the AI system was using for mushroom classification, enabling them to quickly learn how to classify the mushrooms.

The main finding of this study is the interaction effect between explanations and team composition. Although explanations played a key role in influencing decision accuracy in both individual and group settings, their effect on reliance patterns was more significant in the *Group Condition*. When explanations were absent, groups demonstrated higher levels of over-reliance compared to individuals (see Figure 5). This suggests that in instances where the initial decision deviated from the AI recommendation and the AI was

incorrect, groups tended to rely more heavily on the AI system [RQ2].

One possible explanation for this effect is the amplification of cognitive biases such as the anchoring bias [7, 43, 61] and automation bias [35] in group dynamics. The anchoring bias, asserting that individuals tend to rely heavily on the first piece of information they receive, can manifest with the AI recommendation acting as an anchor [61]. In group settings, this effect is magnified, as the AI recommendation becomes an anchor for multiple participants. The qualitative exploratory analysis revealed that participants in the *Group without XAI* condition more frequently cited the AI as their primary influence in making the final decision compared to those in the *Individual without XAI* condition. Another possible reason for this behavior is the diffusion of responsibility within groups, which might reduce the perceived need for critical analysis [8, 15]. In contrast, individuals, bearing sole responsibility, may feel greater pressure to scrutinize decisions carefully and independently. For example, one participant doing the experiment alone without explanations said: *"I tried to figure out what characteristics poisonous mushrooms had in common. I rethought the decisions I made, if I was still confident with my decision, if I wasn't sure I would choose the AI option."*(P. 9).

Moreover, in a group setting, when a recommendation is provided without an accompanying explanation, members may attempt to construct their own mental model to justify the recommendation. This process can ultimately make the group more inclined to follow the suggested direction. Due to phenomena such as groupthink [24], it often takes just one member adopting this reasoning to influence the rest of the group, reinforcing a collective shift toward the recommendation. Additionally, participants may view the AI recommendation as a third element in the group, helping break ties in decision-making and giving more weight to the AI's suggestion. For example, one participant that was in the group conditions without access to explanations said:

"We treated [the AI] like another human and just like a democracy the majority wins."(P. T5)

In contrast, over-reliance rates were significantly lower when explanations were present in groups, even surpassing the over-reliance rates observed in individual conditions. Not only did explanations demonstrate the ability to mitigate the over-reliance rate in group settings, but also to counteract effects seen in individual conditions [RQ2]. The presence of explanations during group discussions potentially facilitated a shared understanding, reducing cognitive load and helping participants understand both the AI system and the provided explanations. This mitigation of over-reliance could be attributed to participants gaining a better understanding of the classification task through the explanations, which enables them to more easily disregard the initial recommendation. This observation was reflected in the qualitative results. Participants in the *Group condition* who had access to explanations mentioned leveraging the AI as a learning resource and frequently relied on their own judgment alongside the AI's input, rather than using the AI as the primary decision-maker. Moreover, an analysis of reliance patterns within the group condition reveals that groups spending more time on their final decision exhibited a lower over-reliance rate. Additionally, groups with access to explanations tended to

take more time deliberating. This suggests that the availability of explanations played a critical role in promoting reflection, leading to longer decision-making processes, and in turn, reducing over-reliance. Explanations encouraged groups to transition from fast, heuristic decision-making to a slower, more reflective approach [25], enabling members to collaboratively clarify misunderstandings or discuss misalignments in their decisions. This dynamic is not accessible to individuals working alone, even when explanations are provided.

These findings underscore the amplified impact of explanations on AI-assisted decision-making within group settings. Explanations are not only essential for reducing instances of over-reliance, but they also play a more significant role in group decision-making than in individual AI-assisted decision-making.

Regarding under-reliance, our study revealed that explanations did not influence under-reliance rates, both in group or individual conditions. In situations where the initial decision differed from the AI recommendation, and the AI was correct, explanations did not contribute to mitigate cases where subjects chose not to rely on the AI system, regardless of whether they were in a group or acting individually. This suggests that addressing over-reliance is comparatively easier than mitigating under-reliance with explanations [RQ2]. Participants in their open-ended questions reported to change to AI's recommendation, only if they were not confident in their initial decision. Example of participant's answer on how to integrate AI recommendations in the individual condition with explanations:

"I tried to learn with the AI[...]. If my initial decision and the one provided by the AI matched, I would keep the same, if they were different I would check why and make my decision based on how sure I was on my previous choice." (P.X8)

This implies that explanations are not enough to trust more in the AI than in oneself, even when AI is correct. The cognitive biases associated with under-reliance patterns, such as algorithm aversion [17] and over-confidence bias, appear to be more robust and influential, making them more challenging to counteract.

Interestingly, our findings are similar to those reported by Chiang et al. [12] in the absence of explanations — the specific context they studied. They observed evidence of increased AI reliance in the *Group* condition, driven by both an increase in over-reliance and a decrease in under-reliance. When examining our results for the *Without XAI* condition, we also observe a tendency for higher reliance and over-reliance rates, along with a lower under-reliance rate in the group setting. However, in our case, these tendencies for reliance and under-reliance are not statistically significant⁸ (see Fig. 5). In the *With XAI* condition, where explanations are present, we observe an inverted trend. The reliance and over-reliance rates are lower for group settings compared to individuals, while the under-reliance rate is slightly higher — although this difference is also not statistically significant.

5.3 Implications for real world scenarios

This research demonstrates how the explainability of AI recommendations can influence group decision-making in practical scenarios.

⁸It is important to note that reliance in their study is measured differently, we argue that the method used in our study is more robust.

In vital group decision-making settings such as Multi-Disciplinary Meetings, where multiple healthcare professionals work together to decide on a patient's diagnosis, prognosis, or treatment [55], or citizen jury deliberations, where a group of citizens discusses and deliberates on an issue within their purview [62], AI tools are already considered integral to the group [13, 44]. When these groups receive AI-generated recommendations, providing explanations for these recommendations not only promotes critical thinking and reduces over-reliance but also prevents exacerbation of reliance issues due to group dynamics in the absence of explanations. While earlier studies have indicated that people tend to prioritize accuracy over explainability in AI-assisted tools during group deliberations on healthcare contexts [62], explanations are crucial in group decision-making to mitigate group biases. These insights highlight the importance of explanations in facilitating well-informed decisions.

The significance of providing explanations extends beyond ensuring system transparency of decision-making mechanisms. This becomes even more evident in group settings, particularly when tasks are performed by lay users [45], who have personal decision-makers, at home, at their fingertips [54]. Our results show that when explanations are offered the interactive nature of group dynamics offers a distinct advantage: the social interaction allows them to challenge AI recommendations and assumptions and improve the quality of their decisions. Therefore, in group decision-making settings AI recommendations have to be accompanied by explanations. In such settings, the lack of clear and effective explanations can exacerbate reliance imbalances and amplify group biases.

To replicate the reflective benefits of group settings for individuals, AI systems could be designed to provide interactive, reflective experiences. For example, incorporating guiding questions or feedback loops that allow users to compare previous decisions or recommendations can prompt deeper thinking and encourage them to reconsider their choices. These mechanisms could help reduce over-reliance on the system by fostering critical engagement with the decision-making process. By embedding such features, individuals could experience a *slower* decision-making process similar to that of groups, leading to more informed decisions and potentially mitigating over-reliance.

6 Limitations and Future directions

The data presented in this paper indicates that explanations positively impact AI-assisted decision-making in group tasks. However, it is important to note that the participants were laypersons involved in the mushroom edibility task, and further research is needed to determine whether these effects are also verified in other contexts with different levels of complexity and risk (e.g. groups of healthcare professionals). Moreover, participants were not informed of the AI model's accuracy beforehand. This design choice was intended to replicate real-world scenarios where the accuracy of an AI model is unknown. As a result, the findings cannot be generalized to scenarios where the accuracy of the AI system is explicitly provided upfront. Providing such information in advance could significantly influence reliance on the AI system.

Additionally, it is worth noting that our study focused solely on feature importance explanations. Different types of explanations

have been shown to yield varied effects, and feature importance explanations, in particular, can elicit multiple responses in terms of trust and reliance [14, 16]. Therefore, future studies could benefit from investigating the generalizability of our results across different types of explanations and varying levels of participant expertise. Moreover, this study focused on 2-person groups. The effect of explanations on groups of decision makers with more than two individuals might be different and should be studied in a posterior study.

In this study, we explored the effects of explanations on group decisions in AI-assisted decision-making. Future research could investigate various aspects of group dynamics, including diverse team compositions, hierarchical structures, and differing levels of expertise among team members.

7 Conclusion

In conclusion, this paper conducts a quantitative experiment exploring the impact of explanations in AI-assisted decision-making within groups. The findings highlight the significant role of XAI in this context, demonstrating its potential in reducing over-reliance in groups, while the absence of explanations tends to increase over-reliance rates. Although the impact of explanations on addressing under-reliance appears limited, these insights offer valuable contributions to our understanding of group dynamics in AI-assisted decision-making. This study represents a pioneering effort in examining how explanations influence group decisions and open opportunities for future research in the human-AI interaction community.

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